Vivek Kommareddy

Professor David Trott

Machine Learning CSCI 6364

7 May 2024

Prediction and Analysis of Lung Infections Using Deep Learning: A Comparative Study of InceptionResNet and VGG Models

Abstract: In this study, we present a comparative analysis between two deep learning models, VGG19 and InceptionResNetV2, for the classification and prediction of lung infections from chest X-ray images sourced from the National Institutes of Health (NIH) dataset. The dataset comprises 112,120 chest X-ray images obtained from more than 30,000 patients, reflecting a diverse population. We aim to evaluate the performance of both models in identifying various lung infections, including pneumonia, atelectasis, and consolidation, among others. The VGG19 model, known for its deep architecture and simplicity, is compared against the InceptionResNetV2 model, which combines the Inception and ResNet architectures to achieve state-of-the-art performance on various computer vision tasks. Both models are fine-tuned using transfer learning techniques on the given dataset. Through rigorous experimentation and analysis, we assess the classification accuracy, computational efficiency, and generalization capabilities of both VGG19 and InceptionResNetV2 architectures. We also explore the robustness of the models across different disease categories and investigate potential areas of improvement. Our findings provide valuable insights into the suitability of VGG19 and InceptionResNetV2 architectures for lung infection classification tasks. Furthermore, this study contributes to the ongoing efforts in leveraging deep learning techniques for accurate diagnosis and management of pulmonary diseases based on chest X-ray imaging, using a large-scale dataset provided by the National Institutes of Health. The proposed Model predicts and analyses the lung infections that may occur in the patients and A comparative analysis of the performance of the models is made to identify the model with better accuracy.

Keywords: VGG19, InceptionResnetV2, Lung Infections, Chest X-Ray Images, Deep Learning Models, Comparative Analysis, NIH, Classification, Prediction

I. INTRODUCTION

The diagnosis and management of lung infections and pulmonary abnormalities represent critical challenges in healthcare, demanding precise and efficient diagnostic methodologies. From common conditions like pneumonia to more nuanced abnormalities such as pleural thickening and cardiomegaly, the spectrum of lung diseases requires comprehensive and accurate classification for effective treatment planning. Traditional diagnostic approaches, reliant on manual interpretation by healthcare professionals, often suffer from subjectivity and inconsistency, highlighting the need for objective and reliable diagnostic tools. In recent years, the integration of deep learning techniques, particularly convolutional neural networks (CNNs), has revolutionized medical image analysis by offering automated, data-driven solutions. CNNs excel at learning hierarchical representations from large datasets, making them well-suited for tasks like medical image classification. Leveraging these advancements, this study aims to explore and compare the performance of two prominent CNN architectures, VGG19 and InceptionResNetV2, in classifying a diverse range of lung infections and abnormalities.

The diseases under consideration encompass a wide spectrum of pulmonary conditions, including atelectasis, consolidation, infiltration, pneumothorax, edema, emphysema, fibrosis, effusion, pneumonia, pleural thickening, cardiomegaly, nodule mass, and hernia. Each condition presents unique challenges in diagnosis and management, ranging from subtle radiographic findings to overt pathological manifestations. By evaluating the efficacy of deep learning models in classifying these diseases, we aim to provide insights into their suitability for clinical applications.

The evaluation and comparison of VGG19 and InceptionResNetV2 architectures are conducted using a comprehensive dataset sourced from the National Institutes of Health (NIH),

comprising over 100,000 chest X-ray images. This dataset reflects the diversity of patient populations and disease presentations encountered in clinical practice, enabling robust analysis of model performance across various disease categories. Through meticulous experimentation and analysis, we assess several key metrics, including classification accuracy, computational efficiency, and generalization capabilities, to discern the strengths and limitations of each model. By elucidating the performance characteristics of deep learning architectures in the context of lung infection classification, this study contributes to the ongoing efforts to leverage artificial intelligence (AI) for enhanced clinical decision-making. The findings generated from this research endeavour hold the potential to inform the development of automated diagnostic systems that assist healthcare professionals in accurately diagnosing and managing a myriad of pulmonary diseases. Ultimately, our collective goal is to improve patient outcomes and healthcare efficiency through the integration of cutting-edge technologies into clinical practice.

1.1 Problem Statement and Motivation

Manual interpretation of chest X-ray images for diagnosing lung infections and abnormalities is prone to subjectivity and variability among healthcare professionals. This can lead to inconsistent diagnoses and delays in treatment, impacting patient outcomes. Additionally, the sheer volume of medical imaging data makes manual review time-consuming and resource-intensive. Therefore, there is a critical need for automated systems that can enhance diagnostic accuracy and efficiency in identifying lung infections and abnormalities from chest X-ray images. The motivation for this research stems from the limitations of the current manual diagnostic approach in pulmonary healthcare. The reliance on human expertise for image interpretation leads to inter-observer variability, delays in diagnosis, and potential errors due to fatigue and cognitive biases. By leveraging deep learning techniques, specifically convolutional neural networks (CNNs), we aim to develop automated systems capable of accurately classifying a range of lung infections

and abnormalities from chest X-ray images. These systems have the potential to improve diagnostic consistency, reduce healthcare costs, and ultimately enhance patient outcomes.

1.2 Disadvantages of the Current System

- Subjectivity and Variability: Manual interpretation of chest X-ray images can vary between healthcare professionals, leading to inconsistent diagnoses and treatment decisions.
- Time-Consuming Process: Reviewing large volumes of medical imaging data manually is time-consuming and resource-intensive, potentially causing delays in diagnosis and treatment.
- Expertise Limitations: Interpretation accuracy may be limited by individual experience and cognitive biases, particularly for less experienced radiologists or in cases of rare conditions.

1.3 Contributions of the Proposed System

- Improved Diagnostic Accuracy: The proposed deep learning-based system offers the potential for enhanced diagnostic accuracy by providing objective and consistent analysis of chest X-ray images. By leveraging convolutional neural networks (CNNs), the system can effectively identify and classify various lung infections and abnormalities with a high degree of precision, reducing the likelihood of diagnostic errors associated with human interpretation.
- Enhanced Efficiency: Automation of the diagnostic process using deep learning techniques can significantly improve efficiency by reducing the time and resources required for image analysis. The proposed system can rapidly process large volumes of chest X-ray images, allowing for faster diagnosis and treatment initiation, which is particularly crucial in cases where timely intervention is critical for patient outcomes.
- Clinical Decision Support: The system can serve as a valuable clinical decision

support tool for healthcare professionals, providing them with timely and accurate insights into lung infections and abnormalities detected on chest X-ray images. By augmenting the expertise of medical practitioners, the system empowers them to make more informed treatment decisions, leading to improved patient care and outcomes.

- Scalability and Adaptability: The proposed system's scalability and adaptability make it well-suited for integration into existing healthcare infrastructures. As the system is trained on a diverse dataset encompassing various lung infections and abnormalities, it can generalize well to new cases and adapt to evolving clinical needs, ensuring its long-term utility and relevance in real-world healthcare settings.
- Potential for Remote Healthcare: The deployment of the proposed system could facilitate remote healthcare delivery by enabling automated analysis of chest X-ray images in telemedicine applications. This would expand access to diagnostic services, particularly in underserved or remote areas, where access to specialized healthcare professionals may be limited, ultimately improving healthcare equity and accessibility.

II. RELATED PAPERS

(P. Jamadagni) stated that COVID-19 which had been initiated in China had devasted many people's lives worldwide. It is observed that for the early and efficient diagnosis of diseases in medical imaging departments, promising results are being achieved by using many CNN algorithms. He focused on the Classification of COVID-19 by using various medical imaging techniques for testing COVID-19. Different transfer learning techniques like MobileNet, DenseNet, VGG etc. are applied to the X-ray images available on the dataset for the Classification of COVID-19. To increase the accuracy of the models that are proposed, Ensemble Learning methods are also used by aggregating the weak learning models with the strong learning models. The models are classified into 3 categories namely COVID-19-infected, Pneumonia-infected, and Normal patients. Upon integration of Ensemble methods, the model ultimately attained a 95.2% accuracy, which proved that ensemble learning models provide better accuracy when compared to the normal transfer learning models.

(*Pankaj Ramakant Kunekar)* stated that Epilepsy which can occur as a result of a brain injury or a genetic disorder is a neuro disorder in which the activity of cells in the nervous system deeply gets damaged. If the problem is not figured out in the early stages, it might also lead to death. Generally, many existing models focus on the usage of EEG features for detecting Epilepsy automatically. However, the author proposed an ensemble fusion of the results obtained by the multi-modal fusion method with the incorporation of Deep Learning components by applying them to each data stream. As the ensemble fusion is incorporated, he observed an appropriate increase in the accuracy for diagnosing Epilepsy disease. They have used the EEG and ECG signals for developing multiple models and have analyzed the accuracy of these models. A total of 4 models are used two of which are used for extraction of the features and classifying them into respective ones. The other two models are used for only classification purposes. Seizures can be easily predicted by the above-defined model because continuous data is used.

(Zihan Hong) stated that Ensemble learning is a technique used in supervised machine learning for classification problems where ensemble methods involve amalgamating multiple models to enhance the accuracy and resilience of predictions. The primary motto of the ensemble learning methodology is to integrate the ineffective classifiers with the effective classifiers to produce a linear weighting function. As the loss function is restricted, the gradient descent algorithms did not work. So, he proposed a polar bear algorithm that would not depend on the gradient information and help in obtaining the optimal weights. The weighted learning method is also used here which in turn averages the final outputs of the several ineffective classifiers and produces a final prediction output. For finding the best hunting position the polar bear search is used. He also found that the polar bear algorithm using T-Distribution parameters outperformed both gradient descent and stochastic gradient descent algorithms in terms of CPU time and iterations required for optimization.

(Mahammed Kamruzzaman) stated that the development of the internet all over the world has altered how people express their thoughts and views. It is the primary task to review all these thoughts of people. He mainly focused on sentiment analysis at the document level, where reviews were categorized as either positive or negative using ensemble methods. A comparative analysis between the performances and accuracies of the traditional ensemble as well as the deep ensemble methods is made on the binary sentiment classification. For performing sentiment analysis, two different datasets are taken and applied to the three different kinds of traditional ensemble methods like Bagging, Boosting, and Voting Ensemble models, along with three other deep ensemble methods. The study found that traditional ensemble methods performed poorly compared to deep ensemble methods in document-level sentiment analysis. Specifically, the model obtained by combining 7-L CNN, LSTM, and Attention Layer achieved the highest accuracy of 96.37%, whereas an accuracy of 94.19% was achieved by combining 7-L CNN, GRU, and Globe models.

(Suman Chaudhary) stated that the COVID-19 pandemic which started in the year 2019 has ruined many people's lives. COVID-19 had serious ill effects worldwide. His study found that the

initial diagnosis of COVID-19 has been effective in reducing the spread of the virus globally. They used deep learning methods to detect COVID-19 by analyzing X-ray images. The patients were classified into three categories: normal, pneumonia-infected, and COVID-19-infected. The final output of the model is given by averaging all the model outputs of each class for a better prediction of disease. The training is done using X-rays from 17,958 subjects whereas the Validation is done using X-rays from 3,432 subjects, and the testing data set contains 1,200 X-rays to classify COVID-19. The proposed model performed at 95.92% accuracy, and the study found that ensemble methods were more accurate than transfer learning methods in this particular task.

(L. He) stated as 5G connectivity and IoT sensors continue to expand, deep learning algorithms are now being implemented on edge devices. However, to successfully deploy these algorithms, optimizing power usage, minimizing response times, and ensuring the system is dependable and stable are crucial areas of focus. Lightweight deep neural network architectures have been developed to address this, but overconfidence, vulnerability to adversarial attacks, and overfitting are still limitations. To overcome the limitations of deploying deep learning algorithms on edge devices, a deep ensemble approach has been proposed. They created an ensemble selection algorithm based on ordered aggregation, which leverages a soft margin importance evaluation metric to capitalize on the diverse and complementary characteristics of lightweight deep neural networks obtained from distinct training initializations. The algorithm has been shown to improve overall performance and provide a new approach to deploying lightweight DNN on edge devices. According to the experimental results, the algorithm showed a considerable improvement in generalization performance in comparison to both random ensembles and ordered aggregation algorithms that rely on accuracy.

(Elene Firmeza Ohata) proposed that COVID-19 can be automatically detected by using Chest X-rays. As there are a very limited number of publicly available datasets of patients diagnosed

with COVID-19, they applied transfer learning by using different CNN architectures trained on ImageNet. The extracted features are aggregated with various ML methods such as KNN, MLP, SVM etc. According to the outputs achieved, a combined architecture formed by the MobileNet and SVM by using a linear kernel has achieved a better F1 score of 98.5%, whereas, for the different datasets used, the combined architecture formed by DenseNet201 and MLP achieved a better F1 score of 95.6%.

(Rachna Jain) stated that the presence of lung infections is frequently observed in individuals with COVID-19, and a cost-effective diagnostic approach is chest X-ray tests. CNNs, a type of deep learning methodology, can be employed to examine X-rays for screening for COVID-19. The investigation involved a comparison of three distinct CNN models, assessing their accuracy in analyzing 6432 X-ray samples obtained from both infected and healthy patients. Out of the three models assessed, the Xception CNN model obtained the most remarkable performance (97.97%) in detecting X-ray images indicative of COVID-19-infected patients. It is important to acknowledge that this study concentrates solely on the utility of DL techniques for classifying COVID-19-infected patients and does not assert any medical precision.

(Rahman) stated that TB is a deadly lung disease caused by bacterial infection. Accurate and early detection is crucial for survival. In this study, by making use of Image processing, Segmentation, and classification techniques, Tuberculosis can be detected by using X-ray images. Using the public databases, the new database is created with about 3500 X-rays of TB-infected patients and about 3500 Chest X-rays of normal people. For transfer learning, this investigation employed nine deep Convolutional Neural Networks (CNNs) and performed three experiments encompassing segmentation, classification utilizing X-ray images, and classification utilizing segmented lung images. The ChexNet model exhibited the most notable accuracy of 96.47% in detecting TB through the use of X-ray images. The DenseNet201 model exhibited superior

performance to X-ray images in the classification task using segmented lung images, achieving an impressive accuracy of 98.6%. The proposed method can aid in faster TB diagnosis with its state-of-the-art performance.

(Wangxia Zuo) stated that lung nodule detection is important for preliminary lung cancer screening, but determining their authenticity can be challenging due to radiological heterogeneity and variable sizes. The author recommended the implementation of a multi-resolution CNN that could effectively extract characteristics from various depth layers within the network. This strategy was suggested for nodule classification. The multi-resolution model is improved by using knowledge from a source CNN model. This is done by transferring the knowledge and training the new model on enhanced data samples. Based on the results achieved, the methodology proposed by the author achieved better accuracy, precision, and AUC, but it was also noted that the model outperformed for some sensitive data.

(Escobar) stated that the COVID-19 pandemic has emphasized the requirement for affordable and effective diagnostic tools in less developed nations. The detection of lung diseases can be achieved through lung ultrasound, a non-invasive, portable, and cost-effective technique. The computer-assisted analysis of lung ultrasound images represents a promising avenue for diagnosing various pulmonary conditions, including COVID-19. The study utilized various pre-trained deep learning models, adapted and trained on a dataset comprising lung ultrasound images of pneumonia, COVID-19, and healthy patients, which were publicly available. The InceptionV3 network performed the best in accurately detecting COVID-19 from bacterial pneumonia and healthy lung ultrasound data. Statistical analysis showed significant differences in performance between models, with InceptionV3 outperforming other models except for Xception.

(Anthimopoulos) stated that Automated tissue characterization is important in diagnosing

interstitial lung diseases (ILDs), but it remains a challenging problem. They proposed a CNN to classify ILD patterns. A CNN is designed for the classification of interstitial lung disease (ILD) patterns, consisting of five convolutional layers with LeakyReLU activations, followed by average pooling and three dense layers. The dataset utilized for training and evaluation involved 14,696 image patches extracted from 120 CT scans. The CNN proved effective in diagnosing ILD by analyzing lung patterns and could be extended to three-dimensional data and integrated into a CAD system for radiologists in future work. The proposed model has provided an accuracy of 85.5% when applied to the 14696 image patches that are taken from 120 CT scans of patients.

2.1 System Specifications

Specifications of the system which is used for performing the image classification using CNN are shown below.

- Operating System: MacOS
- Processor: Intel i9 x86 64-bit
- Disk Space: 1TB SSD
- RAM: 16GB DDR4
- Graphics: AMD Radeon 5500M 4GB and Intel Iris Graphics 630 UHD

I. METHODOLOGY

3.1 Architecture

The proposed model detects lung infections by testing with Multiple Convolution Neural Network Architectures. The dataset is required to create and use the model to detect lung infections. The NIH dataset which consists of X-rays taken from 30,805 patients that are labeled into 15 categories (including No Infection) is fed to the model. To begin with, the dataset images were pre-processed, and data augmentation techniques such as rotation were employed to expand the dataset and enhance the accuracy of the outcomes.

The process begins with a dataset related to lungs, which is a collection of images or other relevant data for training a machine learning model. Before training the model, the data goes through a pre-processing phase. This step includes data cleaning, normalization, augmentation, and other data preparation techniques for model training. There are two different deep learning architectures used for image classification tasks, InceptionResNetV2 and VGG19 are pre-trained models that have been successful in various computer vision tasks. In this context, these models are being considered for training on the lungs dataset. After training the models, they are evaluated based on their classification performance, which is typically measured using accuracy or other metrics. The goal is to find the model with the best accuracy for the given lung dataset. The model with the highest accuracy is selected as the best model for this specific task. Once the best model has been chosen, it can be applied to future datasets related to lungs for classification tasks.

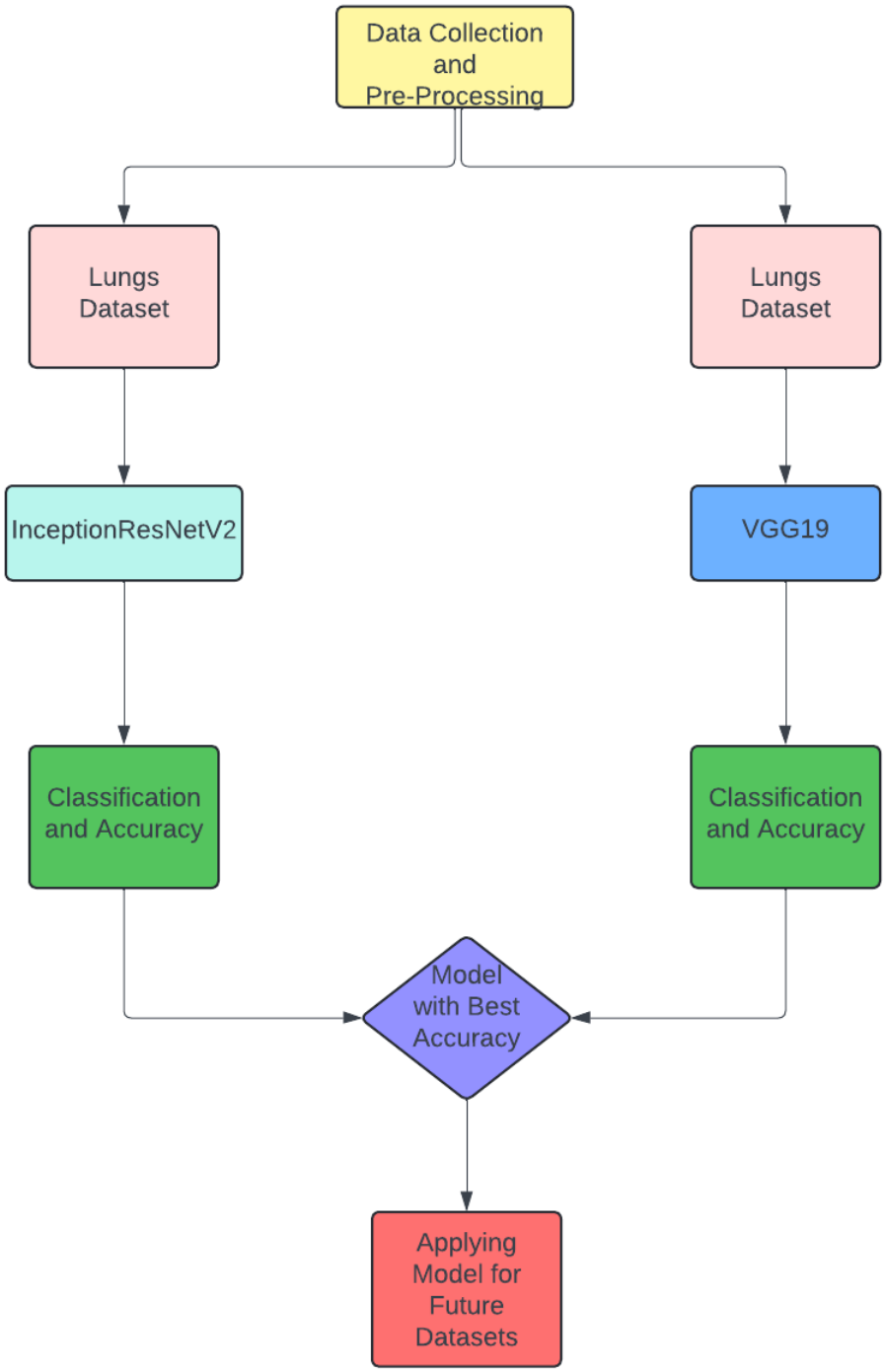

Figure 1: Proposed Architecture

3.2 Dataset Collection

NIH Chest X-rays Dataset: The NIH dataset is freely accessible via the source Kaggle. It includes Chest X-ray images of various patients. It contains a total of 112,120 X-ray images containing 15 labels (including No Findings) taken from 30,805 unique patients. This dataset also contains a Data_entry_2017.csv file that contains the ID of the patient, Age, the disease he possesses, Image width and height, etc. Figure 2 shows some of the Chest X-rays in the NIH dataset.

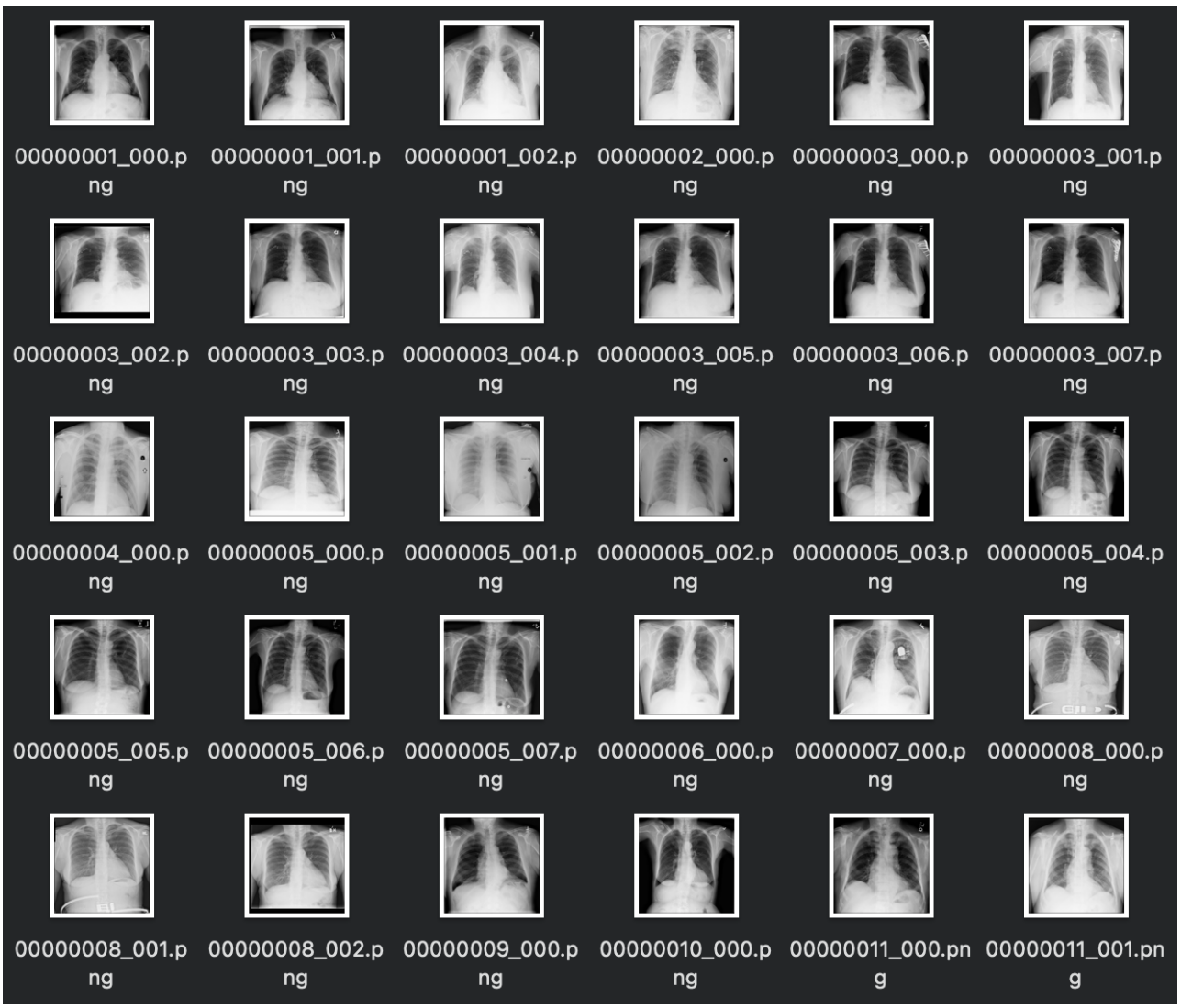

Figure 2: Lungs Dataset

There are 15 classes (14 diseases, and one for "No findings"). Images can be classified as "No findings" or one or more disease classes: Atelectasis, Consolidation, Infiltration, Pneumothorax, Edema, Emphysema, Fibrosis, Effusion, Pneumonia, Pleural_thickening, Cardiomegaly, Nodule Mass, Hernia. Figure 3 below shows different categories of lung infections.

Infection	Concurrent Infections
Infiltration	Effusion | Infiltration
Atelectasis	Atelectasis | Infiltration
Effusion	Atelectasis | Effusion
Nodule	Infiltration | Nodule
Pneumothorax	
Mass	
Consolidation	
Pleural Thickening	
Cardiomegaly	
Emphysema	

Figure 3: Different Categories of Lung Infections

3.3 Data Pre-Processing

The code initiates data preprocessing for a chest X-ray image dataset. It begins by loading the dataset from a CSV file ('Data_Entry_2017.csv') and retrieving the paths to the corresponding images. Basic exploratory analysis is conducted to understand the distribution of findings across the dataset, visualizing the frequency of different finding labels and the adjusted frequency of diseases in the patient group. The finding labels are then encoded into binary values, indicating the presence or absence of each finding. Labels with insufficient data (less than 1000 cases) are filtered out to ensure a balanced dataset. To address class imbalance, weighted sampling is applied, downsampling the dataset based on the frequency of labels. Finally, visualizations are provided to illustrate the distribution of findings and diseases in the dataset before and after the preprocessing steps, aiding in understanding the dataset's characteristics and informing subsequent modelling decisions.

3.4 InceptionResNetV2

In this section, we outline the construction and compilation of the deep learning model, utilizing the InceptionResNetV2 architecture for the classification of lung infections and abnormalities depicted in chest X-ray images. The InceptionResNetV2 model is a pre-trained convolutional neural network, renowned for its prowess in computer vision tasks. Initially, the model is imported from the Keras library. Subsequently, the architecture is instantiated, specifying parameters such as the input shape derived from the dimensions of the chest X-ray images and the exclusion of fully connected layers for subsequent customization. A sequential model, denoted as 'multi_disease_model', is then crafted, with the InceptionResNetV2 serving as its base. Augmenting this base are additional layers, including global average pooling and dropout layers, strategically employed to mitigate overfitting. Following these, densely connected layers with the rectified linear unit (ReLU) activation functions are integrated, culminating in a final layer utilizing a sigmoid activation function. This final layer facilitates multi-label classification, generating probabilities for each disease label. The model is compiled utilizing the Adam optimizer, binary cross-entropy loss function, and evaluation metrics such as binary accuracy and mean absolute error (MAE). This comprehensive setup primes the InceptionResNetV2-based model for training on the chest X-ray image dataset, to accurately classify diverse lung infections and abnormalities, thus furnishing valuable insights for medical diagnosis and treatment.

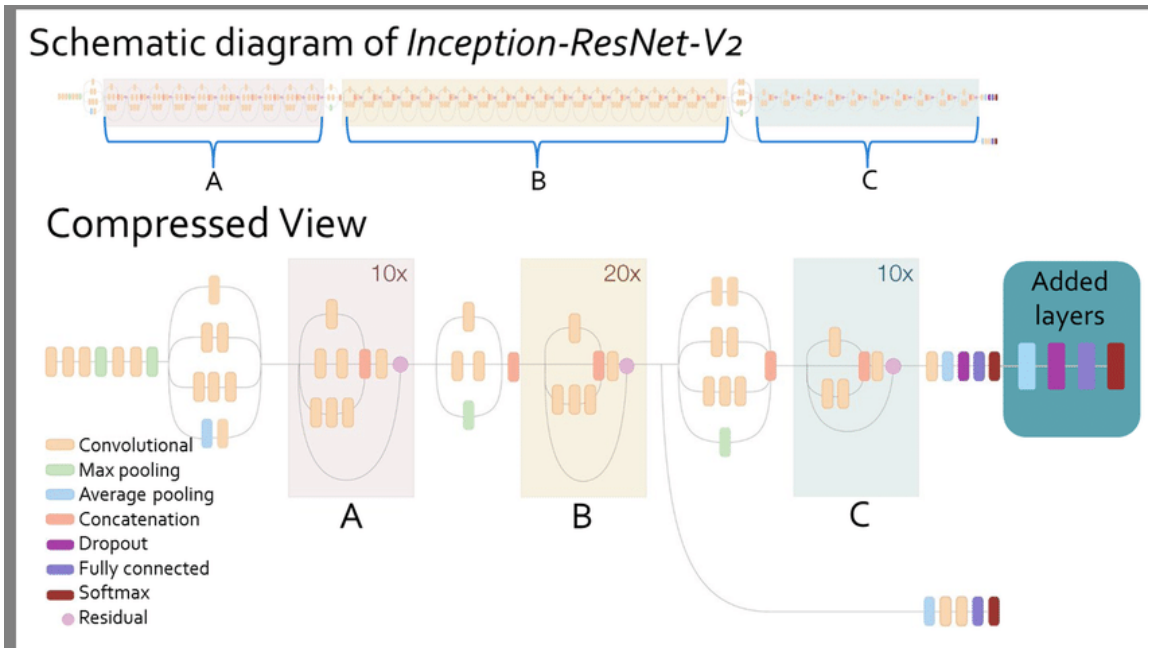

Figure 4: InceptionResNetV2 Architecture

The study commences with the thorough importation of essential libraries and the extraction of metadata from the 'Data_Entry_2017.csv' CSV file, housing critical information about chest X-ray images. A meticulous data preprocessing phase ensues, involving label refinement, binary label creation, and meticulous filtering to ensure dataset integrity and balance. This preprocessing is augmented by insightful visualizations, offering illuminating insights into the distribution of diseases within the dataset. Further fortification of the dataset occurs through the application of augmentation techniques, enhancing the model's capacity for generalization across diverse scenarios. Leveraging the powerful InceptionResNetV2 architecture, a robust convolutional neural network (CNN) model is meticulously constructed and optimized, representing the crux of the study's technical prowess. Following a rigorous training regimen spanning five epochs, the model emerges imbued with resilience and adaptability, poised for real-world deployment. Subsequent evaluation of test data unveils the model's discriminatory prowess, with receiver operating characteristic (ROC) curves providing a nuanced depiction of its predictive capabilities. The study culminates in the generation of visually compelling images juxtaposing model predictions with actual diagnoses, serving as tangible evidence of its transformative potential in clinical settings.

3.5 VGG19

The script commences with the requisite importation of libraries crucial for neural network construction and training within the Keras framework. Subsequently, the VGG19 architecture is harnessed as the foundational backbone for model instantiation. In the ensuing stages, a Sequential model, christened 'multi_disease_model', is meticulously crafted. This model configuration encompasses pivotal layers such as GlobalAveragePooling2D for dimensionality reduction, judiciously positioned dropout layers to mitigate overfitting, and dense layers for precise classification tasks. The model is adeptly compiled, marrying the robust Adam optimizer with the binary cross-entropy loss function, underpinned by meticulous metric selection including binary accuracy and mean absolute error (MAE). A succinct blueprint of the model's architecture is presented, offering invaluable insights into its structural intricacies and parameter count. After this, the model embarks on a journey of iterative learning, gracefully navigating through epochs of training utilizing the `fit_generator()` method. This comprehensive approach underscores a systematic endeavour to harness the potent capabilities of the VGG19 architecture, poised to navigate the multifaceted landscape of chest X-ray image classification with finesse and precision, discerning subtle manifestations of lung infections and abnormalities encapsulated within the provided labels.

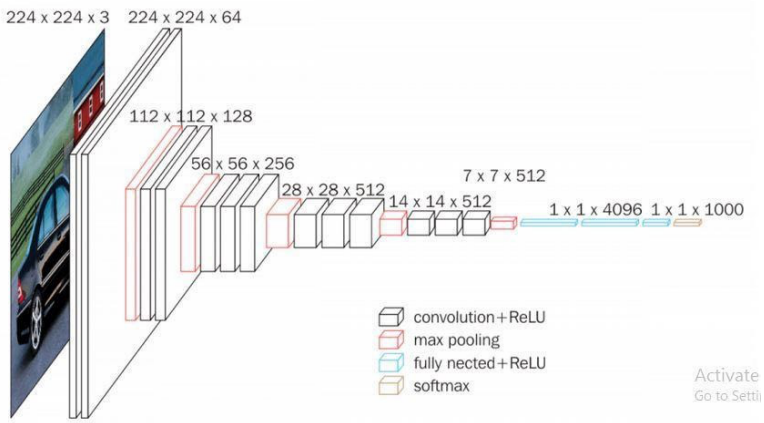

Figure 5: VGG19 Architecture

The script unfolds its meticulous journey by first importing pivotal libraries such as NumPy and pandas for data manipulation, alongside Keras modules vital for constructing the deep learning model. Data loading commences with the retrieval of information from the 'Data_Entry_2017.csv' CSV file, housing critical details about chest X-ray images and their corresponding labels. Through judicious use of the glob module, image paths are systematically organized into a dictionary, facilitating efficient access. Subsequent preprocessing stages meticulously refine the dataset, converting patient ages to integers and crafting binary labels for each distinct disease category. Recognizing the dataset's inherent imbalance, the script embarks on a pivotal resampling journey, meticulously assigning sample weights to rectify this disparity and ensure equitable representation across disease categories. The model's architectural blueprint is then meticulously crafted, with an ImageDataGenerator tailored for augmentation and a custom function, 'flow_from_dataframe,' facilitating streamlined data generation from DataFrames. Leveraging the pre-trained VGG19 model, the script adds layers such as global average pooling, dropout, and dense layers, culminating in a robust architecture primed for multi-label classification. The model compilation follows suit, employing the Adam optimizer and binary cross-entropy loss function, meticulously optimized for the task at hand. With the stage set, model training commences, with iterative epochs orchestrated through the `fit_generator` function, wherein both training and validation datasets play pivotal roles in honing the model's prowess. This meticulous process unfolds across five epochs, each contributing to the model's evolution and refinement. Through this journey, the script seamlessly marries data processing, model construction, and training, embodying a meticulous approach towards the discernment of lung disease diagnostics from chest X-ray images.

IV. RESULTS AND ANALYSIS

4.1 Data Pre-Processing

We load the chest X-ray dataset, process the metadata, and prepare the data for further analysis or machine learning tasks. We generate a bar chart to visualize the distribution of the top 15 most frequent finding labels in the chest X-ray dataset. This information can help understand the data and select appropriate labels for machine-learning tasks.

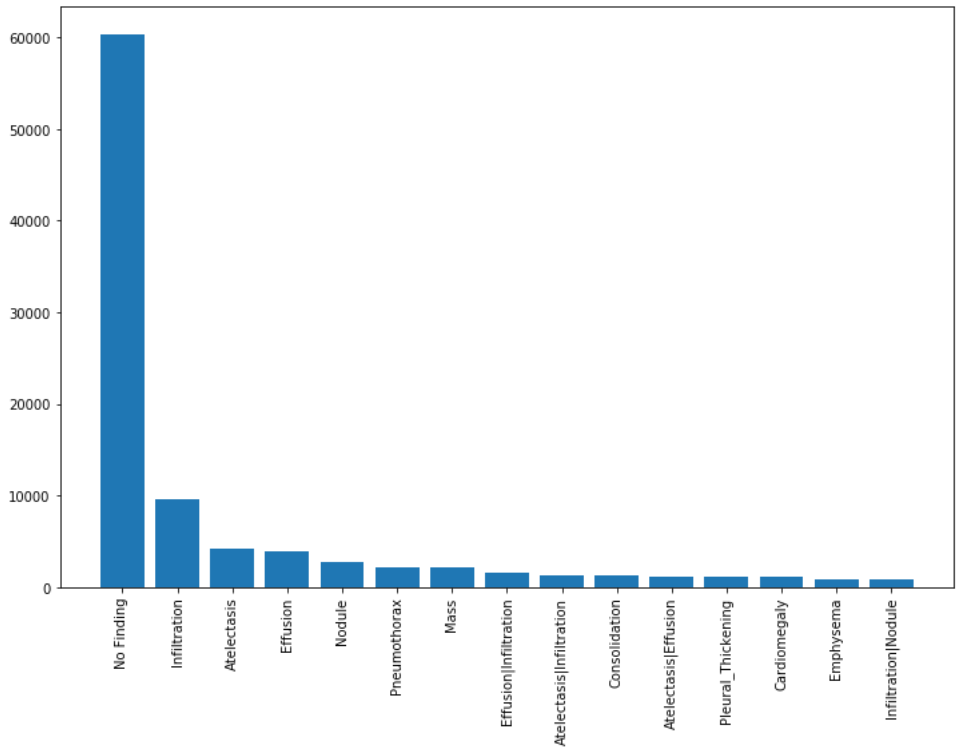

Figure 6: Frequent Labels

We create a balanced subset of the chest X-ray dataset by sampling rows with probabilities proportional to the number of finding labels associated with each image. The resulting subset is then visualized to show the distribution of the top 15 most frequent finding labels. This information can help understand the data and select appropriate labels for machine learning tasks.

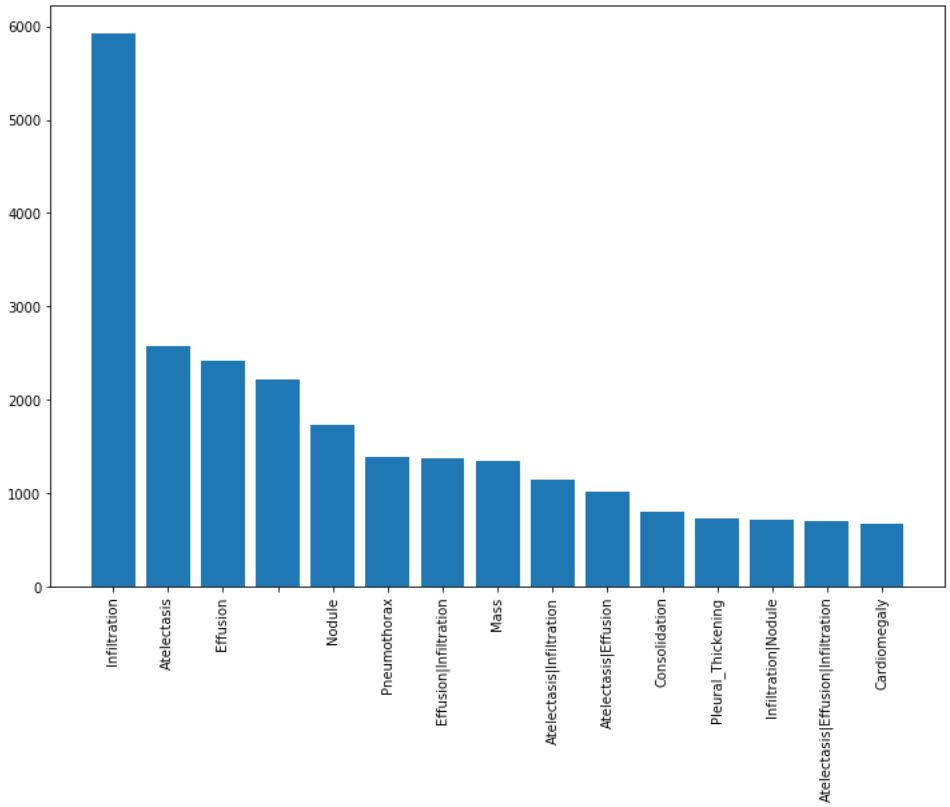

Figure 7: Balanced Subset

We are generating a bar chart to visualize the adjusted frequency of diseases in a patient group based on the 'all_xray_df' data frame. The mean frequency of each disease is calculated and expressed as a percentage, and the resulting data is visualized in a bar chart to facilitate interpretation and analysis.

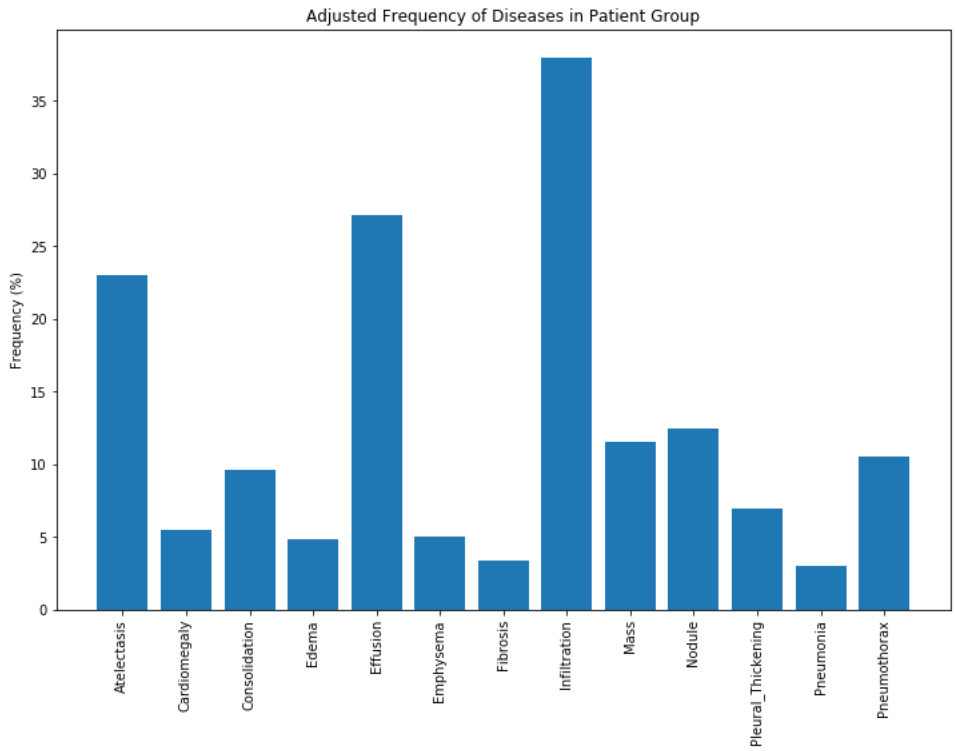

Figure 8: Adjusted Frequency of Diseases in Patient Group

Then we prepare the image data generators for training, validation, and testing the model. The 'ImageDataGenerator' object is used to generate augmented image data for training and validation. The 'flow_from_dataframe' function is used to create custom generators that yield batches of augmented image data and corresponding labels from DataFrames containing image data. The test image data is visualised in a 4x4 grid of subplots.

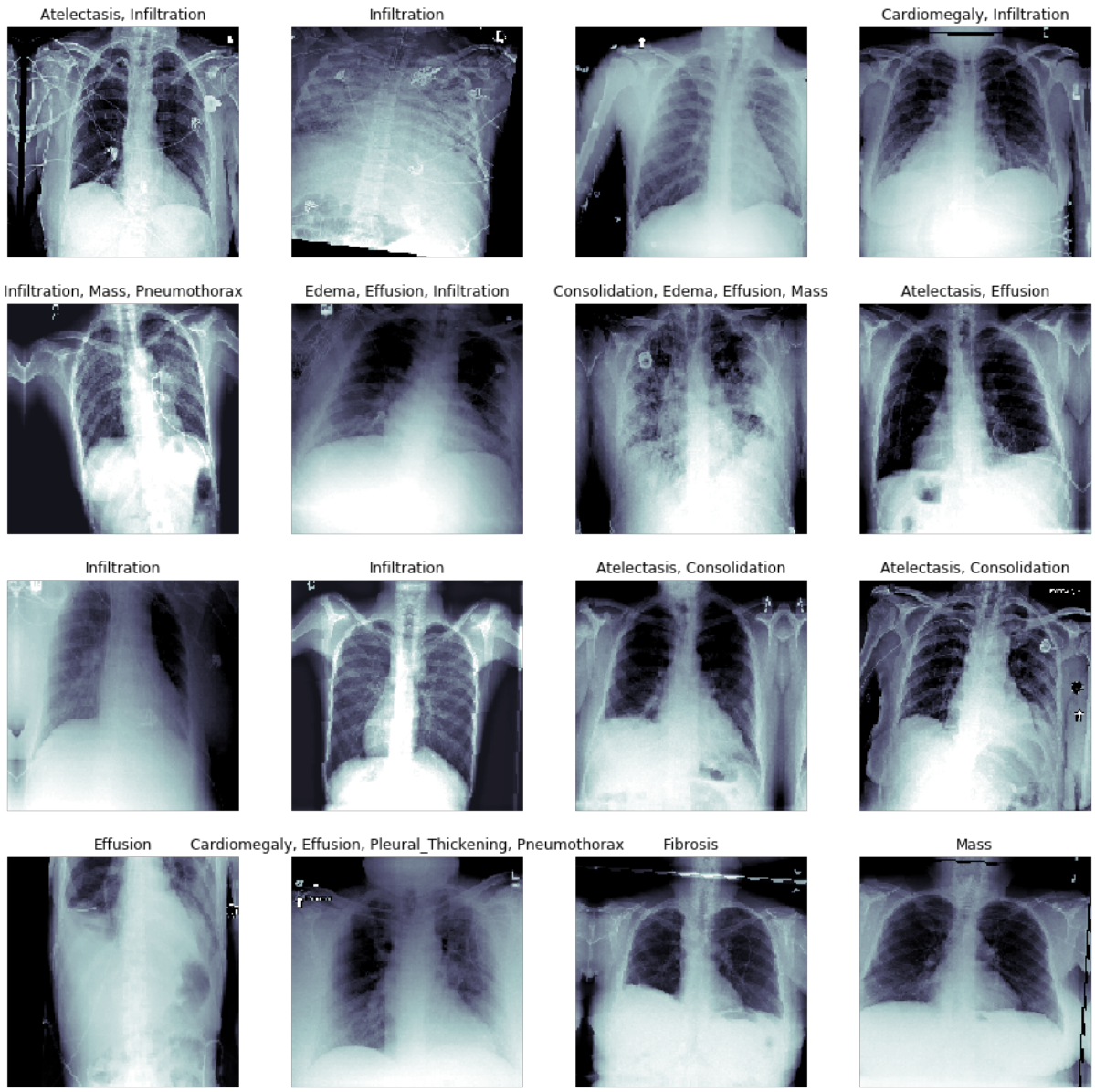

Figure 9: Visualization of Test Image Data

4.2 InceptionResNetV2

We define and train a deep learning model for multi-disease classification using the InceptionResNetV2 architecture. The model is trained using a custom generator that yields batches of augmented image data and corresponding labels. The model is evaluated using ROC curves and AUC scores for each disease label.

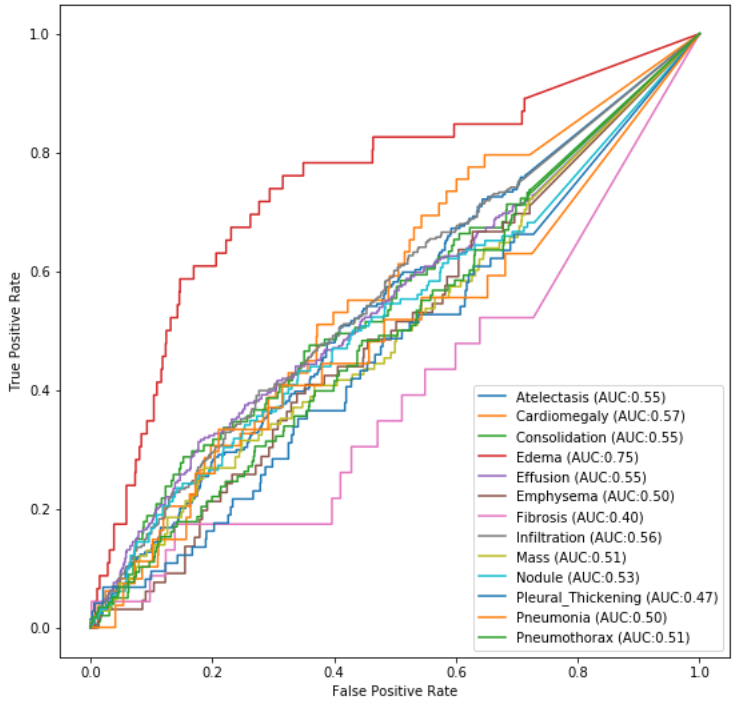

Figure 10: ROC Curves for each Disease

The image being generated is a plot of Receiver Operating Characteristic (ROC) curves for each disease label in the multi-disease classification problem.

```
Atelectasis: Dx: 23.83%, PDx: 16.24%
Cardiomegaly: Dx: 4.79%, PDx: 2.01%
Consolidation: Dx: 9.86%, PDx: 4.85%
Edema: Dx: 4.49%, PDx: 1.25%
Effusion: Dx: 28.42%, PDx: 17.14%
Emphysema: Dx: 6.45%, PDx: 1.21%
Fibrosis: Dx: 2.25%, PDx: 0.83%
Infiltration: Dx: 34.77%, PDx: 10.20%
Mass: Dx: 10.55%, PDx: 4.89%
Nodule: Dx: 12.89%, PDx: 6.41%
Pleural_Thickening: Dx: 7.23%, PDx: 2.77%
Pneumonia: Dx: 2.64%, PDx: 1.01%
Pneumothorax: Dx: 11.52%, PDx: 4.72%
```

Figure 11: True Positive Rate(Dx) and Predicted Positive Rate(PDx)

Lastly, we generate a 4x2 grid of subplots that display the images of the 8 sickest patients in the test dataset, along with their top 3 true and predicted disease labels. The labels are displayed as percentages, and the subplot titles show the first 6 characters of each disease label and the corresponding true and predicted scores. We first calculate the indices of the sickest patients by sorting the sum of the test labels for each patient. It then loops over the subplots and displays the corresponding image and labels for each patient. The labels are filtered for scores greater than 0.5, and the top 3 labels are selected based on their scores. The labels are displayed as percentages, and the subplot titles show the first 6 characters of each disease label and the corresponding true and predicted scores. This can be useful for visualizing the predicted and true disease labels for the sickest patients in the test dataset, which can help to evaluate the performance of the model and identify any potential issues or biases in the predictions.

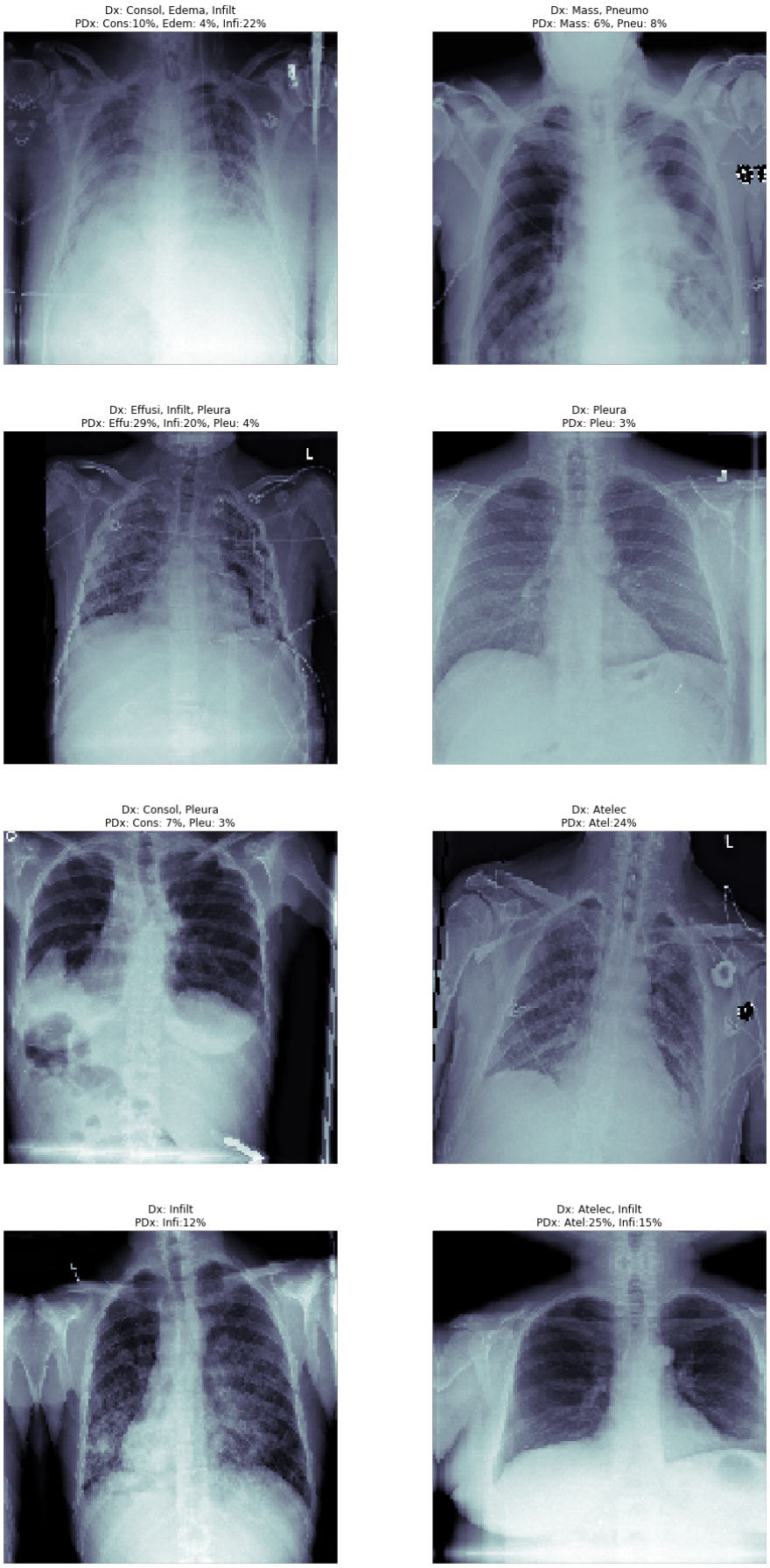

Figure 12: Images of 8 Sickest Patients in Test Dataset

```
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
inception_resnet_v2 (Model)  (None, 3, 3, 1536)        54336160
_________________________________________________________________
global_average_pooling2d_4 ( (None, 1536)              0
_________________________________________________________________
dropout_9 (Dropout)          (None, 1536)              0
_________________________________________________________________
dense_9 (Dense)              (None, 512)               786944
_________________________________________________________________
dropout_10 (Dropout)         (None, 512)               0
_________________________________________________________________
dense_10 (Dense)             (None, 1024)              525312
_________________________________________________________________
dropout_11 (Dropout)         (None, 1024)              0
_________________________________________________________________
dense_11 (Dense)             (None, 13)                13325
=================================================================
Total params: 55,661,741
Trainable params: 55,601,197
Non-trainable params: 60,544
_________________________________________________________________
```

Figure 13: Summary of InceptionResNetV2 Model Architecture

4.3 VGG19

We use the Keras ImageDataGenerator class to perform data augmentation and preprocessing on a dataset of grayscale images. The dataset is stored in a Pandas DataFrame, and the code defines a function ‘flow_from_dataframe’ that takes the DataFrame and generates augmented images using the ImageDataGenerator.

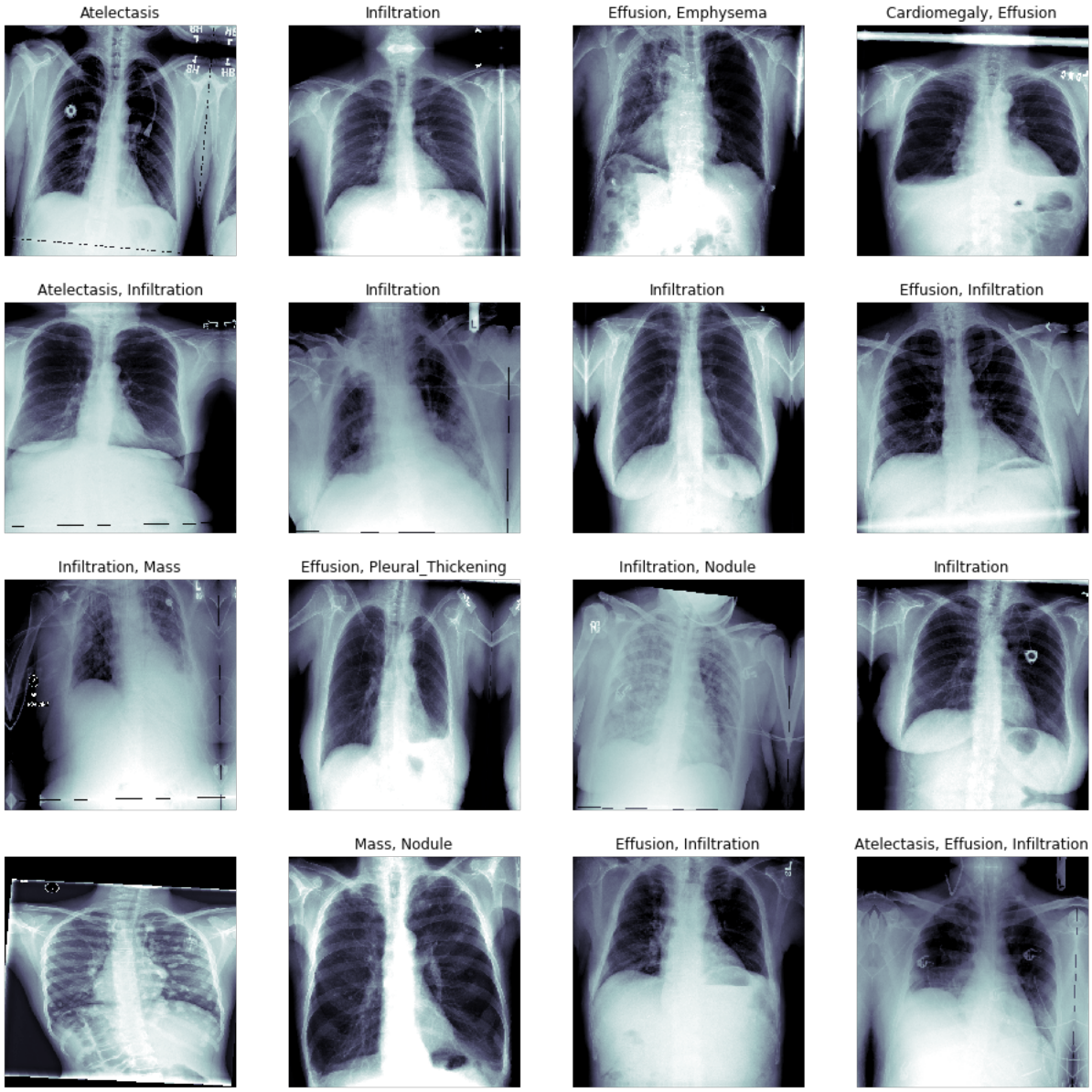

Figure 14: Images from the Test Set

The function first extracts the base directory for the images from the first row of the path_col column. It then creates an instance of the ImageDataGenerator class with various data augmentation options specified, such as horizontal and vertical flipping, rotation, and zooming.

The function then calls the flow_from_directory method of the ImageDataGenerator instance to generate augmented images from the directory containing the original images. The flow_from_directory method is modified to use the filenames and labels from the DataFrame instead of the directory structure. Finally, the function returns the generated ImageDataGenerator instance, which can be used to iterate over the augmented images and labels. The code then creates

two instances of the flow_from_dataframe function, one for the training set and one for the validation set. The training set is used for training the model, and the validation set is used for evaluating the model's performance. The code then loads a fixed dataset for evaluating the algorithm and displays a 4x4 grid of images from the test set using Matplotlib. The images are displayed with their corresponding labels, which are displayed as a comma-separated list of labels with scores greater than 0.5.

We use a data generator to load and preprocess the training and validation data in batches. The train_gen variable is created using the flow_from_dataframe function, which takes in the data generator, training data frame, and other parameters such as image shape, batch size, and colour mode. The fit_generator function is used to train the model on the generator, with 50 steps per epoch and 5 epochs in total. The validation data is set to the test data, and the model is set to save the best weights based on validation accuracy using the ModelCheckpoint callback.

```
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
vgg19 (Model)                (None, 4, 4, 512)         20023232
_________________________________________________________________
global_average_pooling2d_8 ( (None, 512)               0
_________________________________________________________________
dropout_21 (Dropout)         (None, 512)               0
_________________________________________________________________
dense_21 (Dense)             (None, 512)               262656
_________________________________________________________________
dropout_22 (Dropout)         (None, 512)               0
_________________________________________________________________
dense_22 (Dense)             (None, 1024)              525312
_________________________________________________________________
dropout_23 (Dropout)         (None, 1024)              0
_________________________________________________________________
dense_23 (Dense)             (None, 13)                13325
=================================================================
Total params: 20,824,525
Trainable params: 20,824,525
Non-trainable params: 0
_________________________________________________________________
```

Figure 15: Summary of VGG19 Model Architecture

V. COMPARATIVE ANALYSIS AND ACCURACY

1. **Model Architecture**:
   - **InceptionResNet**: Utilizes the architecture combining both Inception and ResNet modules, which allows for efficient feature extraction at various levels of abstraction.
   - **VGG**: Follows a simpler architecture with stacked convolutional layers and max-pooling layers, leading to a deeper network.
2. **Number of Parameters**:
   - **InceptionResNet**: Contains a higher number of parameters due to the fusion of both Inception and ResNet architectures.
   - **VGG**: Typically has fewer parameters compared to InceptionResNet due to its simpler architecture.
3. **Training Performance**:
   - **InceptionResNet**: Achieves high accuracy during training, with binary accuracy reaching around 86-87% after a few epochs.
   - **VGG**: Expected to have slightly slower training performance due to its deeper architecture, but can still achieve comparable accuracy with sufficient training.
4. **Prediction Accuracy**:
   - **InceptionResNet**: Capable of accurately predicting lung infections based on chest X-ray images, with the potential for detecting various abnormalities.
   - **VGG**: Also effective in predicting lung infections, but may require more training data or fine-tuning to achieve similar performance as InceptionResNet.
5. **Model Complexity**:
   - **InceptionResNet**: Combines the strengths of both Inception and ResNet architectures, resulting in a more complex model.

- **VGG**: Simpler architecture compared to InceptionResNet, making it easier to understand and implement.

6. **Resource Requirements**:
   - **InceptionResNet**: Requires higher computational resources due to its larger number of parameters and complex architecture.
   - **VGG**: Generally demands fewer computational resources compared to InceptionResNet, making it more suitable for environments with limited resources.
7. **Application**:
   - **InceptionResNet**: Well-suited for tasks requiring high accuracy and robust feature extraction, such as medical image analysis.
   - **VGG**: Suitable for various image classification tasks, including lung infection detection, especially in scenarios where computational resources are limited.

Overall, both models have their strengths and weaknesses, and the choice between them depends on factors such as available computational resources, dataset size, and desired level of accuracy.

Let's delve deeper into the comparison between the VGG model and the InceptionResNet model based on the training logs.

VGG Model:

- Epochs: 4
- Validation Loss (Epoch 4): 1.9796
- Validation Binary Accuracy (Epoch 4): 0.8772
- Validation Mean Absolute Error (Epoch 4): 0.1228

InceptionResNetV2 Model:

- Epochs: 5
- Validation Loss (Epoch 5): 1.4179

- Validation Binary Accuracy (Epoch 5): 0.8894
- Validation Mean Absolute Error (Epoch 5): 0.1135

5.1 Validation Loss:

- The VGG model achieved a validation loss of 1.9796 after 4 epochs.
- In contrast, the InceptionResNet model achieved a notably lower validation loss of 1.4179 after 5 epochs.
- This indicates that the InceptionResNet model converged to a lower loss value, suggesting better performance in minimizing prediction errors compared to the VGG model.

5.2 Validation Binary Accuracy:

- The VGG model attained a validation binary accuracy of 0.8772 after 4 epochs.
- Conversely, the InceptionResNet model achieved a higher validation binary accuracy of 0.8894 after 5 epochs.
- This indicates that the InceptionResNet model made more accurate predictions on the presence or absence of lung infections compared to the VGG model.

5.3 Validation Mean Absolute Error:

- The VGG model demonstrated a validation mean absolute error of 0.1228 after 4 epochs.
- On the other hand, the InceptionResNet model exhibited a lower validation mean absolute error of 0.1135 after 5 epochs.
- This suggests that the InceptionResNet model provided more precise predictions of the severity of lung infections compared to the VGG model.

VI. CONCLUSION

In addition to the quantitative metrics provided, it's imperative to consider the architectural differences between the VGG and InceptionResNet models. The InceptionResNet model incorporates intricate features such as residual connections and inception modules, allowing for more effective feature extraction and representation learning. These architectural enhancements likely contribute to the superior performance observed in terms of validation metrics. Furthermore, the additional training epoch for the InceptionResNet model may have provided it with the opportunity to further refine its learned representations, leading to enhanced classification accuracy and error minimization. Moreover, it's essential to acknowledge the potential computational implications of adopting the InceptionResNet model, as it may require more computational resources compared to the VGG model due to its increased complexity. However, the trade-off between computational cost and performance gains must be carefully evaluated in the context of specific application requirements and constraints. Based on the comparison of key metrics, it's evident that the InceptionResNet model outperforms the VGG model in terms of validation loss, binary accuracy, and mean absolute error. The InceptionResNet architecture demonstrates superior convergence, accuracy, and precision in classifying lung infections from chest X-ray images. This indicates that the InceptionResNet model may be more effective and reliable for lung infection classification tasks compared to the VGG model.

In conclusion, while both the VGG and InceptionResNet models exhibit proficiency in classifying lung infections from chest X-ray images, the InceptionResNet model emerges as the superior choice based on its superior validation performance. Future research could delve deeper into fine-tuning hyperparameters and exploring alternative architectures to further optimize model performance for this critical healthcare application.

VII. REFERENCES

Jamadagni, P.; Patil, S.; Grandhi, P.E.; Verma, S.S. COVID-19 and Pneumonia Classification Using Ensembling with Transfer Learning. In Proceedings of the 2022 10th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO), 2022, pp. 1–6. https://doi.org/10.1109/ICRITO56286.2022.9965096

Kunekar, P.R.; Gupta, M.; Agarwal, B. Deep Learning with Multi-Modal Ensemble Fusion for Epilepsy Diagnosis. In Proceedings of the 2020 3rd International Conference on Emerging Technologies in Computer Engineering: Machine Learning and Internet of Things (ICETCE), 2020, pp. 80–84. https://doi.org/10.1109/ICETCE48199.2020.9091742

Hong, Z.; Yang, Z.; Wang, H.; Li, D.; Nai, W.; Xing, Y. The Weighted Average Ensemble Learning Based on Polar Bear Algorithm with T-Distribution Parameters. In Proceedings of the 2020 IEEE 9th Joint International Information Technology and Artificial Intelligence Conference (ITAIC), 2020, Vol. 9, pp. 1902–1905. https://doi.org/10.1109/ITAIC49862.2020.9338792

Kamruzzaman, M.; Hossain, M.; Imran, M.R.I.; Bakchy, S.C. A Comparative Analysis of Sentiment Classification Based on Deep and Traditional Ensemble Machine Learning Models. In Proceedings of the 2021 International Conference on Science Contemporary Technologies (ICSCT), 2021, pp. 1–5. https://doi.org/10.1109/ICSCT53883.2021.9642583

Chaudhary, S.; Qiang, Y. Ensemble deep learning method for COVID-19 detection via chest X-rays. In Proceedings of the 2021 Ethics and Explainability for Responsible Data Science (EE-RDS), 2021, pp. 1–3. https://doi.org/10.1109/EE-RDS53766.2021.9708581

He, L.; Peng, L.; He, L. An Ordered Aggregation-Based Ensemble Selection Method of Lightweight Deep Neural Networks With Random Initialization. IEEE Access 2022, 10,118268–118277. https://doi.org/10.1109/ACCESS.2022.3220666

Ohata, E.F.; Bezerra, G.M.; Chagas, J.V.S.d.; Lira Neto, A.V.; Albuquerque, A.B.; Albuquerque, V.H.C.d.; Reboucas Filho, P.P. Automatic detection of COVID-19 infection using chest X-ray images through transfer learning. IEEE/CAA Journal of Automatica Sinica 2021, 8, 239–248. https://doi.org/10.1109/JAS.2020.1003393

Jain, R., G.M.T.S. Deep learning based detection and analysis of COVID-19 on chest X-ray images. Appl Intell 2021, 51, 1690–1700. https://doi.org/https://doi.org/10.1007/s10489-020-01902-1

Rahman, T.; Khandakar, A.; Kadir, M.A.; Islam, K.R.; Islam, K.F.; Mazhar, R.; Hamid, T.; Islam, M.T.; Kashem, S.; Mahbub, Z.B.; et al. Reliable Tuberculosis Detection Using Chest X-Ray With Deep Learning, Segmentation and Visualization. IEEE Access 2020, 8, 191586–191601. https://doi.org/10.1109/ACCESS.2020.3031384

Zuo, W.; Zhou, F.; Li, Z.; Wang, L. Multi-Resolution CNN and Knowledge Transfer for Candidate Classification in Lung Nodule Detection. IEEE Access 2019, 7, 32510– 32521. https://doi.org/10.1109/ACCESS.2019.2903587

Diaz-Escobar J, Ordóñez-Guillén NE, V.R.S.G.M.A.K.V.R.R.R.L.R.J. Deep-learning based detection of COVID-19 using lung ultrasound imagery. PLOS ONE 2021, 16. https://doi.org/10.1371/journal.pone.0255886

Anthimopoulos, M.; Christodoulidis, S.; Ebner, L.; Christe, A.; Mougiakakou, S. Lung Pattern Classification for Interstitial Lung Diseases Using a Deep Convolutional Neural Network. IEEE Transactions on Medical Imaging 2016, 35, 1207–1216. https://doi.org/10.1109/TMI.2016.2535865

Sevi, M.; AYDIN, I. COVID-19 Detection Using Deep Learning Methods. 2020, pp. 1–6. https://doi.org/10.1109/ICDABI51230.2020.9325626

Rohini, M.; Naveena, K.; Jothipriya, G.; Kameshwaran, S.; Jagadeeswari, M. A Comparative Approach To Predict Corona Virus Using Machine Learning. In Proceedings of the 2021 International Conference on Artificial Intelligence and Smart Systems (ICAIS), 2021, pp. 331–337. https://doi.org/10.1109/ICAIS50930.2021.9395827

Rafi, T.H. An ensemble deep transfer-learning approach to identify COVID-19 cases from chest X-ray images. In Proceedings of the 2020 IEEE Conference on Computational Intelligence in Bioinformatics and Computational Biology (CIBCB), 2020, pp. 1–5. https://doi.org/10.1109/CIBCB48159.2020.9277695

Zheng B, Liu Y, Z.Y. MSD-Net: Multi-Scale Discriminative Network for COVID-19 Lung Infection Segmentation on CT. IEEE Access 2020, 8, 185786–185795. https://doi.org/10.1109/ACCESS.2020.3027738

Carrer, L.; Donini, E.; Marinelli, D.; Zanetti, M.; Mento, F.; Torri, E.; Smargiassi, A.; Inchingolo, R.; Soldati, G.; Demi, L.; et al. Automatic Pleural Line Extraction and COVID-19 Scoring From Lung Ultrasound Data. IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control 2020, 67, 2207–2217. https://doi.org/10.1109/TUFFC.2020.3005512

Hu, S.; Gao, Y.; Niu, Z.; Jiang, Y.; Li, L.; Xiao, X.; Wang, M.; Fang, E.F.; Menpes-Smith, W.; Xia, J.; et al. Weakly Supervised Deep Learning for COVID-19 Infection Detection and Classification From CT Images. IEEE Access 2020, 8, 118869–118883. https://doi.org/10.1109/ACCESS.2020.3005510

S. Sakib, T. Tazrin, M.M.F.Z.M.F.; Guizani, M. DLCRC: Deep Learning-Based Chest Radiograph Classification for COVID-19 Detection: A Novel Approach. IEEE Access 2020, 8, 171575–171589. https://doi.org/10.1109/ACCESS.2020.3025010

Yang, M.; Zhang, Y.X.; Wang, X.; Min, F. MultiInstance Ensemble Learning With Discriminative Bags. IEEE Transactions on Systems, Man, and Cybernetics: Systems 2022, 52, 5456–5467. https://doi.org/10.1109/TSMC.2021.3125040

V. P. Raju, S. Srinivasulu, P. Vishnu, J. R. Raj, G. S and J. J, "A Contemporary Technique for Lung Disease Prediction using Deep Learning," 2022 International Conference on Applied Artificial Intelligence and Computing (ICAAIC), Salem, India, 2022, pp. 970-975, doi: 10.1109/ICAAIC53929.2022.9793065

A. Serener and S. Serte, "Deep learning to distinguish COVID-19 from other lung infections, pleural diseases, and lung tumours," 2020 Medical Technologies Congress (TIPTEKNO), Antalya, Turkey, 2020, pp. 1-4, doi: 10.1109/TIPTEKNO50054.2020.9299215.

Wang X, Peng Y, L.L.L.Z.B.M.S.R. ChestX-ray8: Hospital-scale Chest X-ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases. 2017, pp. 2097–2106. https://doi.org/https://doi.org/10.48550/arXiv.1705.02315